

Towards an Interoperable Framework for Solar and Heliospheric CubeSat Missions



SOC Integration and Data Standardization across HENON, SEE and CUSP Missions at ASI

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ASI Italian Space Agency

1. INTRODUCTION

This contribution presents the ongoing effort led by the Italian Space Agency (ASI) to establish a standardized interoperability framework between the Science Operations Centers (SOC) of three upcoming CubeSat missions dedicated to the study of the Sun and the Heliosphere. Given the inherent heterogeneity of small-satellite payloads and ground segments, the primary objective is to mitigate data fragmentation and streamline operational workflows. The work focuses on the definition of common data formats, synchronized telemetry processing pipelines, and shared API protocols to ensure that scientific data are not only compatible but seamlessly integrable across different platforms. By implementing these interoperability standards, ASI aims to transform individual mission outputs into a synergistic multi-point observation network, significantly reducing the time-to-science and enhancing the overall reliability of heliospheric data analysis.

2. THE ASSETS

Three CubeSats, One Data Ecosystem

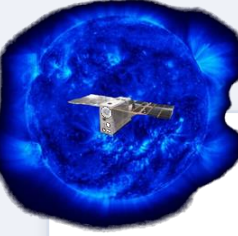


HENON

Programme: Selected through ASI ALCOR Program and developed through ESA General Support Technology Programme (GSTP) – “Fly” Element. Main contrib. IT

Consortium: Argotec (prime, IT), INAF, Aboa Space Research Oy, University of Turku, Imperial College London, Charles University, SpaceDys, IMT srl, University of Calabria, University of Florence, Nautilus, Mars Space Ltd, Techline Systems, AVS UK

Space Weather measurements in Distant Retrograde Orbit (DRO) for 3-6 hour advanced warning of solar storms (sunward side).

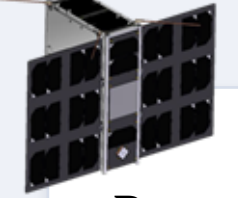


SEE

Programme: ASI ALCOR

Consortium: Argotec (prime, IT), UniToV (IT), INFN (IT), School of Aerospace Engineering (IT), OPTEC (IT), and NEXT (IT).

Full-disk UV imaging at Mg II 280 nm, plus X-ray and gamma-ray light curves at kilohertz cadence.



CUSP

Programme: ASI ALCOR

Consortium: INAF (Prime, IT), University of Bologna (IT), Università della Tuscia (IT), IMT (IT), SCAI Connect (IT)

Hard X-ray polarimetry of solar flares, 25–100 keV — a measurement no current mission routinely provides.

3. THE APPROACH: PLACING DATA AT THE CORE OF MISSION DEVELOPMENT

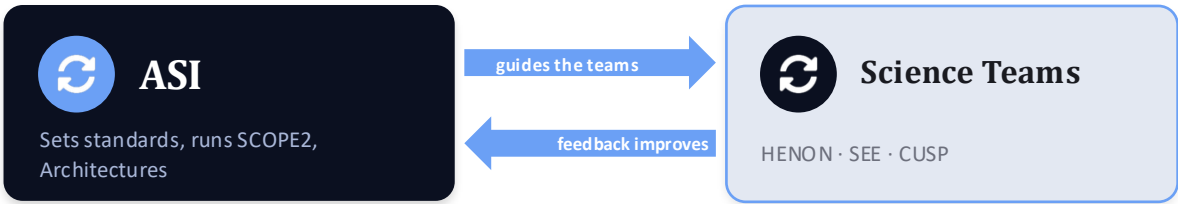
Small-satellite payloads and ground segments are inherently heterogeneous: without a common foundation, scientific data risks staying fragmented across missions.

Driven by the conviction that mission success relies on data accessibility and their interoperability, the work focuses on integration of data models and data formats, harmonization of science plans, ASI is leading the effort to place data at the centre of the development of the three missions.

While observation and operations profiles are mission-specific, substantial value lies in unifying data and science management. Data engineering is therefore embedded throughout all project lifecycles. Under this framework, ASI leads the drive toward **standardized interoperability across SOC architectures, data models and science planning harmonization.**

Based on a **continuous-learning approach, operational feedback and expertise gained alongside science teams are directly channeled into formal guidelines, infrastructure enhancements, and evolving the SCOPE2 platform.**

CONTINUOUS LEARNING APPROACH



4. INFRASTRUCTURE

The ASI Space Science Data Centre



The ASI Space Science Data Center (SSDC) is a dedicated research and operations facility that coordinates scientific mission operations while conducting advanced R&D in space data management and exploitation, in close collaboration with leading Italian research institutes. Spanning diverse domains—from astrophysics and planetary science to heliophysics, particle physics, and life sciences—the center ensures operational continuity, FAIR data interoperability, and long-term scientific exploitation.

SSDC supports mission teams in the design and implementation of the Science Operations Centres (SOC), the engineering of scientific pipelines, and the development of advanced data exploitation tools. To deliver these capabilities, the center relies on its own dedicated high-performance computing and storage infrastructure.

5. WHERE THE WORK STANDS

- Interoperability is a mission requirement, not a downstream repair.** It is an iterative process ASI undertakes together with the HENON, SEE and CUSP science teams, throughout all development phases — refining the framework at each mission step. Progress is reviewed jointly with the SOC teams at every project milestone. The three missions are themselves at different maturity stages — HENON in implementation (Phase D), SEE entering Phase C, CUSP having just closed Phase B — which is exactly why the framework has to absorb asynchronous progress rather than assume a synchronized start.
- Shared infrastructure**

All three missions target the SSDC archive and the SCOPE framework instead of mission-specific data systems — reusing what ASI already operates rather than building three times over.
 - Integrated data model**

Metadata dictionary, product levels and calibration conventions are being written with the SOC teams with interoperability in mind.
 - Native coordination**

Coordination among the missions is built-in from the beginning leading to a better data interoperability and possibly exploitation.
 - ADDED VALUE**

Feedback from every step is reflected into improvements to the platform, procedures, and constitutes an added value for future missions in the broader space science ecosystem.
- 6. IMPACT**
- By implementing these interoperability standards and setting this procedural approach to the development of ground segments and science operation centres, ASI aims to transform individual mission outputs into a synergistic multi-point network – paving the way to a data ecosystem which is a natural consequence of implementing a data co-engineering process across all development phases.
- Science missions developed around data*

DATA CO-ENGINEERING ACROSS ALL PHASES — FOUR LEVELS OF INTEROPERABILITY

01	02	03	04
SOC Architecture	Harmonized Data models	Harmonized Science Operations	Harmonized Science Plans
Shared infrastructure: the SSDC archive and the ASI SCOPE² multi-mission framework — one operational environment instead of three mission-specific systems.			

7. FROM THREE MISSIONS TO ONE DATA ECOSYSTEM

A key expected benefit of this approach is the creation of a seamless and interoperable data ecosystem in which scientific information is readily accessible, discoverable, and usable by a broad community throughout the mission lifetime. By implementing data co-engineering practices and harmonizing science operation modes from the earliest phases of a space mission’s development, data products, metadata standards, calibration procedures, and analysis pipelines can be designed consistently across instruments and mission phases. This reduces fragmentation, minimizes the need for costly post-launch adaptations, and ensures that observations acquired by different instruments can be integrated and analyzed within a common framework.

Such an approach significantly enhances the scientific return of the mission by enabling rapid access to high-quality, well-documented, and interoperable datasets. Researchers can more easily combine complementary measurements, perform multi-instrument and multi-scale investigations, and exploit advanced analysis techniques, including machine learning and data-driven discovery methods. Early harmonization also facilitates long-term data preservation, reproducibility of scientific results, and compatibility with existing international archives and virtual observatory infrastructures. Ultimately, the co-engineering of data systems and science operations promotes a more efficient exploitation of mission resources, accelerates scientific discoveries, and maximizes the impact of the mission for both the heliophysics community and the broader space science ecosystem.